GATE

Computer Science & Information Technology

Computer Networks

Transport Layer

DPP 01

Q1 UDP provides ______ services.
(A) Connection Less' and 'Unreliable'
(B) 'Connection Less' and 'Reliable'
(C) 'Connection Oriented' and 'Unreliable'
(D) 'Connection Oriented' and 'Reliable'

Q2 Consider total length field value of UDP packet header is 200 (in decimal) calcualte its payload size (in bytes)?

Q3 Which transport layer protocol is used for faster communication?
(A) TCP
(B) UDP
(C) IP
(D) None of the above

Q4 Which transport layer protocol is used by HTTP/1.1 (version 1.1)?
(A) TCP
(B) UDP
(C) IP
(D) None of the above

Q5 Identify INCORRECT statement(s) regarding TCP and UDP.
(A) Flow ccontrol provided by both TCP and UDP
(B) Flow control provided by TCP, but not UDP
(C) Flow control provided by UDP, but not TCP
(D) Flow control provided by neither TCP nor UDP

Q6 Consider the 3-way handshake protocol for TCP connection establishment between hosts S and D. Host S sends a TCP connection request message (TCP segment) to D, and D accepts the connection request and send accept message (TCP segment) to S. Identify correct TCP header flag bits in accept message (TCP segment)?
(A) SYN bit = 0 and ACK bit = 0
(B) SYN bit = 0 and ACK bit = 1
(C) SYN bit = 1 and ACK bit = 0
(D) SYN bit = 1 and ACK bit = 1

Q7 Consider a TCP sender sends one segment in which sequence number field value (in decimal) is 2500 and the segment carried 700 bytes of data in its payload. What should be the acknowledgment number sent by receiver to acknowledge this segment successfully?

Q8 Consider the 3-way handshake protocol for TCP connection establishment between hosts S and D. Host S sends a TCP connection request message (TCP segment) to D. D accepts the connection request and send accept message (TCP segment) to S and waiting for ACK of it from S. Identify the current TCP state of Host D?
(A) LISTEN
(B) SYN_SENT
(C) SYN_RECV
(D) ESTABLISHED

Q9 Consider a TCP sender send two segments to TCP receiver one after another. The sequence number field value in the header of the segments (in decimal) are 7500 and 8800 respectively. How many bytes carried by the first segment in its payload field?

Q10 Consider TCP protocol runs over a 300 Mbps network. What should be maximum segment lifetime (in seconds, rounded off to nearest intetger)?

Q11 Consider new reliable byte-stream transport protocol like TCP, runs over a 200 Mbps network and maximum segment lifetime is 600 seconds. Calculate minimum number of bits required for sequence number field in the header?

Q12 Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. The TCP server send ACK to acknowledges the FIN segment, in which state does the server side TCP connection wait to close the connection?

(A) CLOSE-WAIT
(B) TIME-WAIT
(C) FIN-WAINT-1
(D) FIN-WAINT-2

Q13 Which of the following socket API function is/are executed by TCP server?

(A) bind()
(B) connect()
(C) listen()
(D) accept()

Q14 Consider size of congestion window of a TCP connection be 16 MSS when a timeout occurs. The round trip time of the connection is 3 seconds. The time taken (in sec) by the TCP connection to reach up to 12 MSS congestion window is ________.

Q15 Consider in TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm, where the window size at the start of the slow start phase is 1 MSS and the threshold at the start of the first transmission is 32 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size (in MSS) at the end of the tenth transmission?

Answer Key

Q1	(A)
Q2	192~192
Q3	(B)
Q4	(A)
Q5	(A, C, D)
Q6	(D)
Q7	3200~3200
Q8	(C)
Q9	1300~1300
Q10	114~115
Q11	34~34
Q12	(A)
Q13	(A, C, D)
Q14	21~24
Q15	10~10

Hints & Solutions

Q2 Text Solution:

UDP Segment size = Total length = 200 bytes

UDP Header size = 8 byte

UDP Segment Payload size = (200 – 8) bytes = 192 bytes

Q3 Text Solution:

TCP = UDP + Extra Services

UDP = Basic Protocol

IP = Network Protocol

Q5 Text Solution:

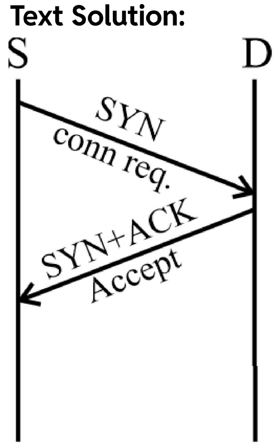
Flow ccontrol provided by both TCP and UDP → False

Flow control provided by TCP, but not UDP → True

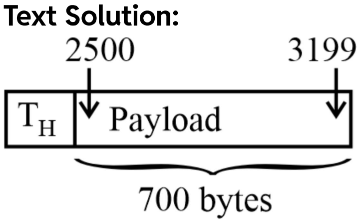
Flow control provided by UDP, but not TCP → False

Flow control provided by neither TCP nor UDP → False

Q6 Text Solution:



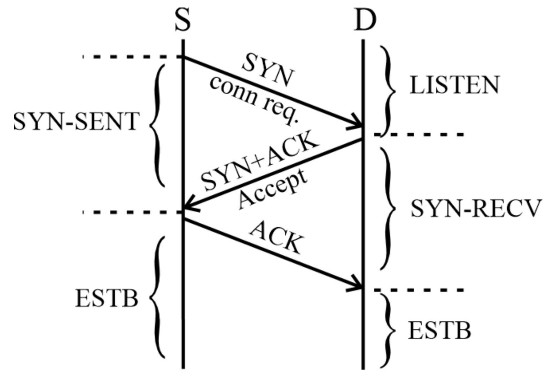
Q7 Text Solution:



Seq. no. = 2500

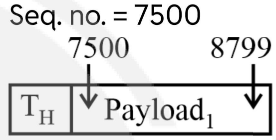
Ack No. = 2500 + 700 = 3200

Q8 Text Solution:

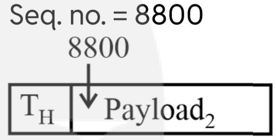


Q9 Text Solution:

Seq. no. = 7500



Seq. no. = 8800



$Payload_1$ size = (8800 – 7500) bytes = 1300 bytes

Q10 Text Solution:

$$\text{MSL} = \frac{2^{32}\ \text{bytes}}{\text{Bandwidth}} = \frac{2^{32}\times 2^{3}\ \text{bits}}{300\ \text{Mbps}}$$

$$= \frac{2^{35}\ \text{bits}}{3\times 10^{8}\ \text{bits / sec}} = 114.53\ \text{sec}$$

Q11 Text Solution:

Maximum no. of bytes can be transmitted in one MSL

= MSL × Bandwidth = 600 sec × 200 Mbps

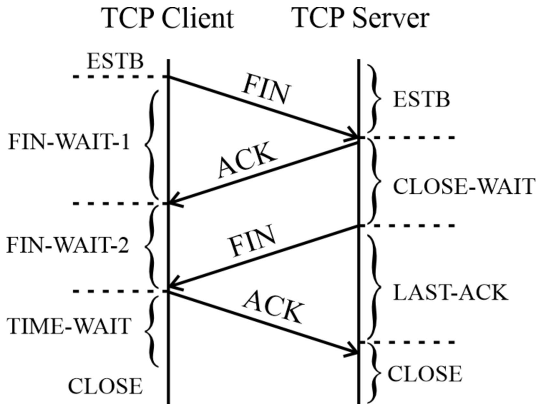
= 12×10^{10} bits $= \frac{12\times 10^{10}}{8}$ bytes

Minimum no. of bits for seq. no. field

$$= \left\lceil \log_2\left(\frac{12\times 10^{10}}{8}\right)\right\rceil \text{bits} = \lceil 33.80 \rceil \text{ bits}$$

$$= 34 \text{ bits}$$

Q12 Text Solution:



Q13 Text Solution:

conenct() → TCP client

Q14 Text Solution:

CWND = 16 MSS → *

Time-out

$$\text{SSThresh} = \frac{\text{CWND}}{2} = \frac{16\ \text{MSS}}{2} = 8\ \text{MSS}$$

RTT = 3 sec

CWND = 1 MSS
1
2 MSS
2
4 MSS
3
8 MSS
4
9 MSS
5
10 MSS
6
11 MSS
7
12 MSS

Time = 7 × RTT = 7 × 3 sec = 21 sec

Q15 Text Solution:

SSThresh = 32 MSS

CWND = 1 MSS
1
2 MSS
2
4 MSS
3
8 MSS
4
16 MSS
5
Timeout

$$\text{SSThresh} = \frac{\text{CWND}}{2} = \frac{16\ \text{MSS}}{2} = 8\ \text{MSS}$$

CWND = 1 MSS
6
2 MSS
7
4 MSS
8
8 MSS
CWND = 8 MSS
9
9 MSS
10
10 MSS